# 9.2 Two-Way Tables and Probability

Name _____ Period ____ Date _____

Algebra I 9.2 (30-minute plan) · Student sheet

**I can…**

- complete a two-way table.
- find joint and marginal relative frequencies.
- find conditional relative frequencies and use them to compare groups.

**Vocabulary** · fill in the missing word

**Two-way table** — a table that shows counts for two _____ at once

**Joint frequency** — a count _____ the table (one row and one column)

**Marginal frequency** — a total in the Total row or _____

**Joint relative frequency** — a joint frequency ÷ the _____ total

**Marginal relative frequency** — a marginal frequency ÷ the _____ total

**Conditional relative frequency** — a joint frequency ÷ the total of its row or _____

## Part 1 · Notes and examples (fill in each note, then work the example)

**Joint frequencies** are _____ the table; **marginal frequencies** are the _____.

### Example 1

A survey asked 80 students whether they own a pet. Complete the two-way table.

| | Owns a pet | No pet | Total |
|---|---|---|---|
| 9th grade | 24 | ? | 40 |
| 10th grade | ? | 12 | ? |
| Total | 52 | ? | 80 |

**Relative frequencies:** Joint relative frequency = cell ÷ _____ Marginal relative frequency = total ÷ _____

**Conditional relative frequency:** Conditional relative frequency = cell ÷ _____

**"Given":** The word "given" tells me which _____ to divide by.

### Example 2

Use the completed table.

a. What percent of the students are 10th graders who own a pet?

b. What percent of the students own a pet?

| | Owns a pet | No pet | Total |
|---|---|---|---|
| 9th grade | 24 | 16 | 40 |
| 10th grade | 28 | 12 | 40 |
| Total | 52 | 28 | 80 |

## Part 2 · Partner practice

### Partner practice

Complete the two-way table about a new lunch menu.

| | Likes it | Doesn't | Total |
|---|---|---|---|
| **Male** | 18 | ? | 30 |
| **Female** | ? | 8 | ? |
| **Total** | **40** | **?** | **?** |